MAT 3103: Computational Statistics and Probability
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Chapter 2 [Probability and Events]

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Basics concepts of Probability:

- **Experiment:** It is an act that can be repeated under similar conditions.
- **Outcomes:** The results of an experiment are known as outcomes.

  Example: If signals are sent from a station, then the signal may be faded (F) or not faded (N) are the outcomes .
- **Sample space:** A set or collection of all possible outcomes of a random experiment is known as sample space. It is denoted by S.

  Example: In tossing a coin, the possible outcomes are Head (H) and Tail (T). So, the sample space for this random experiment is S = {H, T}.
- **Sample point:** Any of the possible outcomes in a sample space is known as sample point.

  Example: In tossing a coin, Head (H) is a sample point and Tail (T) is another sample point.
- **Event:** Any statement regarding one or more of the outcomes of a sample space recorded from a random experiment is known as event.

  Example: I f two computers, one Samsung (S) and one ACER (A) are allotted to two persons to work at different time periods, then the probable arrangements of the computers will be S : { S S, S A, A S, A A }. At this stage one may be interested to calculate the chance of getting the Samsung and ACER computers by them , then the probable results in favour of the statement are A : { S A , A S }. Here statement A is an event. Again, let us consider the occurrence of getting no Samsung computer, then we can write this statement as B : { A A }. This statement B is also an event.
- **Favorable outcomes:** Number of outcomes in favor of an event is known as favorable outcomes. It is denoted by m ($\leq$ n).

  Example: In the previous example, for event A, m=2 and for B m = 1.
- **Probability:** If a random experiment shows n exhaustive, mutually exclusive and equally likely outcomes and if m ($\leq$ n) outcomes are in favor of an event A, then the probability of an event A is measured by :

$P(A) = \frac{\textbf{Favourable no.of outcomes of event A}}{\textbf{Total no.of outcomes of the random experiment}} = \frac{\mathbf{m}}{\mathbf{n}}$; where, $0 \leq P(A) = \frac{m}{n} \leq 1$

- **Independent Events:** Two events A and B are said to be independent, if and only if P(AB) = P(A) P(B).

Example 1: A fair coin is tossed once. What is the probability that a head will be shown?

Solution: A fair coin means, the probability of observing head (H) equal to the probability of observing tail (T). A coin is tossed once, so the sample space is S = {H, T}. Let us now define the event that head will be shown.
Thus the event set A = {H} and

$$P(A) = \frac{\mathbf{m}}{\mathbf{n}} = \frac{\mathbf{1}}{\mathbf{2}}.$$

Example 2: Two fair coins are tossed once. What is the probability that (a) both coins will show head (b) at least one coin will show tail (c) at most (or at best) one coin will show head, and (d) none of the coins will show head?

Solution: A fair coin means, the probability of observing head (H) equal to the probability of observing tail (T).
Sample space for two coins is S = {HH, HT, TH, TT} and the cardinality is m = 4.

(a) Let us define the event as observing head (H) for both coins , that is, A = {HH}. Thus m = 1 and
$P(A) = \frac{\mathbf{m}}{\mathbf{n}} = \frac{\mathbf{1}}{\mathbf{4}}$
(b) At least one coin will show tail (T), that is, the event set B = {HT, TH, TT} and m = 3. Thus
$P(B) = \frac{\mathbf{m}}{\mathbf{n}} = \frac{\mathbf{3}}{\mathbf{4}}$
(c) At best one coin will show head, that is, C = {HT; TH; TT} and $P(C) = \frac{\mathbf{m}}{\mathbf{n}} = \frac{\mathbf{3}}{\mathbf{4}}$
(d) None of the coins will show head, that is, the event set D = {TT} and $P(D) = \frac{\mathbf{m}}{\mathbf{n}} = \frac{\mathbf{1}}{\mathbf{4}}$

Example 3: Three fair coins are tossed once. What is the probability that (i) at least two coins will show head (ii) at most (or at best) two coins will show head, and (iii) first coin will show head (iv) first and third coin will show head (v) first or third coin will show head (vi) first coin shows head given that third coin shows head?

Solution: A fair coin means, the probability of observing head (H) equal to the probability of observing tail (T).
Sample space for three coins is

S={HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}; n=8

(i) Let A be an event that at least two coins will show head. So, A = {HHH, HHT, HTH, THH}. Thus m = 4 and $P(A) = \frac{\mathbf{m}}{\mathbf{n}} = \frac{\mathbf{4}}{\mathbf{8}} = \frac{\mathbf{1}}{\mathbf{2}}$
(ii) At most two coins will show head (H), that is, the event set B = {HHT, HTH, HTT, THH, THT, TTH, TTT} and m = 7. Thus $P(B)) = \frac{\mathbf{m}}{\mathbf{n}} = \frac{\mathbf{7}}{\mathbf{8}}$

(iii) The first coin will show head, that is, C = {HHH, HHT, HTH, HTT} and

$$P(C) = \frac{m}{n} = \frac{4}{8} = \frac{1}{2}$$

(iv) The first and the third coins will show head, that is, the event set D = {HHH, HTH} and

$$P(D) = \frac{m}{n} = \frac{2}{8} = \frac{1}{4}$$

(v) The probability that the first or the third coin shows head, that is, we want to find P(A∪B).

A={HHH. HHT, HTH, HTT}; m=4
B={HHH, HTH, THH, TTH}; m=4
A∩B={HHH, HTH}; m=2

$$P(A\cup B)=P(A)+P(B)-P(A\cap B)=\frac{4}{8}+\frac{4}{8}-\frac{2}{8}=\frac{3}{4}$$

(vi) First coin shows head given that third coin shows head. Thus we would like to find
P(A given B) = P(A conditional on B) = $P(A|B)=\frac{P(A\cap B)}{P(B)}=\ =\ \frac{2/8}{4/8}=\frac{1}{2}$

Probability When Sampling With/Without Replacement:

Example 4: In a packet there are 7 Samsung and 5 Nokia mobile phone sets. Two sets are drawn one after another **a**)at random **(b)** one by one with replacement, **(c)** one by one without replacement. Find the probability that **(i)** both are Samsung sets **(ii)** one set is Samsung and another one is Nokia.

Solution:

a) at Random,
There are 7+5=12 sets. Two sets can be selected in $^{12}C_2$=66ways=n

i) Let A: both are Samsung sets. A can occur in $^{7}C_2$=21ways=m
P(A)=m/n=21/66=7/22

ii)Let B: one set is Samsung and one is Nokia.
B can occur in $^{7}C_1\times{}^{5}C_1$=35ways=m
P(B)=m/n=35/66

b) with replacement,
The sample space to draw 2 sets one after another is S = {SS, SN, NS, NN},
Where S= Samsung, N=Nokia
i)P(A) = P(SS) = $\frac{^{7}C_1}{^{12}C_1}\times\frac{^{7}C_1}{^{12}C_1}=\frac{49}{144}$
ii) P(B) = P(NS) + P(SN) = $\frac{^{5}C_1}{^{12}C_1}\times\frac{^{7}C_1}{^{12}C_1}+\frac{^{7}C_1}{^{12}C_1}\times\frac{^{5}C_1}{^{12}C_1}=\frac{70}{144}$

c) without replacement,
The sample space to draw 2 sets one after another is S = {SS, SN, NS, NN}.

i) P(A) = P(SS) = $\frac{^{7}C_1}{^{12}C_1}\times\frac{^{6}C_1}{^{11}C_1}=\frac{42}{132}$

ii) P(B) = P(NS) + P(SN) = $\frac{^{7}C_1}{^{12}C_1} \times \frac{^{5}C_1}{^{11}C_1} + \frac{^{5}C_1}{^{12}C_1} \times \frac{^{7}C_1}{^{11}C_1} = \frac{70}{132}$

Example 5: In an office there are 5 computers identified by serial number 1,2,3,4 and 5. Two computers are a)selected for two person to work at different working hours, b)selected for two persons to work at the same time. Find the probability that **(i)** both of them get the same number of computers, **(ii)** sum of the numbers of the computers are 8 or first selected computer bears the number 3 **(iii)** sum of the numbers of the computers are 9 or first selected computer bears the number 2, **(iv)** sum of the numbers of the computers are 6 under the condition that second selected computer bears the number 4. c)Two computers are selected at random. Find the probability that i)the numbers of both computers are odd, ii)sum of the numbers of computer is even.

Solution: a) The sample space of the experiment is

$$\left\{\begin{array}{ccccc} 11 & 21 & 31 & 41 & 51 \\ 12 & 22 & 32 & 42 & 52 \\ 13 & 23 & 33 & 43 & 53 \\ 14 & 24 & 34 & 44 & 54 \\ 15 & 25 & 35 & 45 & 55 \end{array}\right\}$$

We have n=25

(i) Let, **A** be the event that both computers are of same number.

Favorable cases to A, m=6; where, A = {11, 22, 33, 44, 55}

$\therefore$ P (**A**) = $\frac{m}{n} = \frac{5}{25} = \frac{1}{5}$

(ii) Let, **B** be the event that sum of the numbers of the computers is 8
and **C** be the event that first selected computer bears the number 3.

- Favorable cases to B, m=3; where, B = {53,44,35}

Favorable cases to C, m=5; where, C= {31, 32, 33, 34, 35,}

Favorable cases to B$\cap$C, m=1; where, B$\cap$C = {35}

P(B or C) = P(B$\cup$C) = P(B) + P(C) $-$ P(B$\cap$C) = $\frac{3}{25} + \frac{5}{25} - \frac{1}{25} = \frac{7}{25}$

(iii) Let, **D** be the event that sum of the numbers of the computers 9
and **E** be the event that first selected computer bears the number 2.

- Favorable cases to D, m=2; where, D = {45,54}
- Favorable cases to E, m=5; where, E = {21, 22, 23, 24, 25,}
- Favorable cases to D∩E, m=0; so, D & E are mutually exclusive events.

P(D or E) = P(D∪E) = P(D) + P(E) − P(D∩E) = $\frac{2}{25} + \frac{5}{25} - \frac{0}{25} = \frac{7}{25}$

(iv) Let, F be the event that sum of the numbers of the computers 6 and G be the event that first selected computer bears the number 4.

Favorable cases to F, m=5; where, F = {15, 33, 24, 51, 42}
Favorable cases to G, m=5; where, G = {14,24,34,44,54}
Favorable cases to F∩G, m=1; where, F∩G = {24}

$$P(F/G) = \frac{P(F\cap G)}{P(G)} = \frac{1/25}{5/25} = \frac{1}{5} \quad \text{given } P(G)\neq 0$$

P (F/G) is called **Conditional Probability** of F under the condition that G occurs first.

b) The sample space is

S=

	21	31	41	51
12		32	42	52
13	23		43	53
14	24	34		54
15	25	35	45	

Here, N=20

i) A is impossible event, P(A)=0

ii)B={53,35,}, C={31,32,34,35}

P(B or C) = P(B∪C) = P(B) + P(C) − P(B∩C) = $\frac{2}{20} + \frac{4}{20} - \frac{1}{20} = \frac{5}{20}$ =1/4

iii)D={45,54},E={21,23,24,25}

P(D or E) = P(D∪E) = P(D) + P(E) − P(D∩E) = $\frac{2}{20} + \frac{4}{20} - \frac{6}{20}$ = .3

iv)F={15,24,51,42},G={14,24,34,54}

Favorable cases to F∩G, m=1; where, F∩G = {24}

$P(F/G) = \frac{P(F \cap G)}{P(G)} = \frac{1/20}{4/20} = \frac{1}{4}$ given $P(G) \neq 0$

c) The sample space is, S={12,13,14,15,23,24,25,34,25,45}

Two computers can be taken from 5 in $^5C_2$=10ways=n

i)let A: number of both computers are odd, A={13,15,35},m=3

P(A)=m/n=3/10=.3

ii) Let B: sum of number of computers is even, B={13,15,24,35},m=4

P(B)=m/n=4/10=.4

Unequal Probability:

Example 6: 85% e-mails sent from a cyber cafe reach to the destination properly. Once 3 mails are checked randomly, Find the probability that **(i)** all 3 reach properly, **(ii)** two reach properly, **(iii)** at least one reaches properly, **(iv)** at best two reach properly.

Solution: Let, R = reach properly the e-mail and N = not reach properly the e-mail.

Sent of 3 e-mails can occur in $n = 2^3 = 8$ ways. The sample space is-

S = { RRR, RRN, RNR, RNN, NRR, NRN, NNR, NNN }

Given, P(R) = 0.85 and P(N) = 1−0.85 = 0.15. So, R and N are not equally.

(i) Let, **A**: all 3 reach properly; where, A = {RRR}

P(**A**) = P(RRR) = 0.85×0.85×0.85 = 0.614125

(ii) Let, **B**: two reach properly; where, B = {RRN, RNR, NRR}

P(**B**) = P(RRN)+ P(RNR)+ P(NRR)

= (0.85×0.85×0.15)+(0.85×0.15×0.85)+(0.15×0.85×0.85) = 0.325125

(iii) Let, C: at least one reaches properly;

where, C = {RRR, RRN, RNR, RNN, NRR, NRN, NNR} $\bar{C}$ = {NNN}

P(C) =1- P ($\bar{C}$) = 1- P(NNN) = 1- (0.15×0.15×0.15) = 0.996625

(iv) Let, D: at best two reach properly;

where, D = {RRN, RNR, RNN, NRR, NRN, NNR, NNN} $\bar{D}$ = {RRR}

P(D) =1- P ($\bar{D}$) = 1- P(RRR) = 1- (0.85×0.85×0.85) = 0.385875

Equal and Unequal probability in one problem:

Example 7: In a communication system signals are sent, where some of the signals are faded (F) and some reached to the destination properly (P). If power of the signals is not appropriate, then 50% signals are faded. But if signals are sent with sufficient power, still 10% signals are faded. An inspection team observed 3 consecutive signals. Find the probability that i) no signal is faded. ii) One signal is faded. iii) At least one signal is faded. iv) At best one signal is faded.

Solution: The sample space of the inspection is

S= {PPP, PPF, PFF, PFP, FFP, FPP, FPF, FFF}

First case, P (F) =P (P) =1/2

So, equally likely cases are, n=8

i)Let A: no signal is faded, A={PPP},m=1

P (A) =m/n=1/8

ii) Let B: one signal is faded, B= {PPF, PFP, FPP}, m=3

P (B) =m/n=3/8

iii)Let C: At least one signal is faded, C={PPF,PFF,PFP,FFP,FPP,FPF,FFF}, m=7

P(C)=m/n=7/8

[Alternative: here, C=$\bar{A}$, P(C)= P($\bar{A}$)=1-P(A)=1-(1/8)=7/8]

iv)Let D: at best one signal is faded., D={PPP,PPF,PFP,FPP}, m=4

P(D)=m/n=4/8=1/2

Second case, Given P(P)=0.9, P(F)=0.1

i)P(A)= P(PPP)= 0.9×0.9×0.9=.729

ii)P(B)= P(PPF)+P(PFP)+P(FPP)=.9×.9×.1+.9×.1×.9+.1×.9×.9=.273

iii)As C=$\bar{A}$, P(C)=P($\bar{A}$)=1-P(A)=1-.729=.271

iv)P(D)=P(PPP)+P(PPF)+P(PFP)+P(FPP)

=.9×.9×.9+.9×.9×.1+.9×.1×.9+.1×.9×.9=.972

Probability from Outcome Matrix:

Example 8: In an office there are 50 computers, out of which 20 are ACER and 30 are Samsung. The computers are investigated and found that 15 ACER and 25 Samsung computers are good. One computer is selected at random. Find the probability that the selected computer is **(i)** Either Samsung or good compter. **(ii)** ACER given that it is good.

Solution: Let, A: Samsung computer, $\bar{A}$: ACER computer

G: Good, $\bar{G}$: Not Good

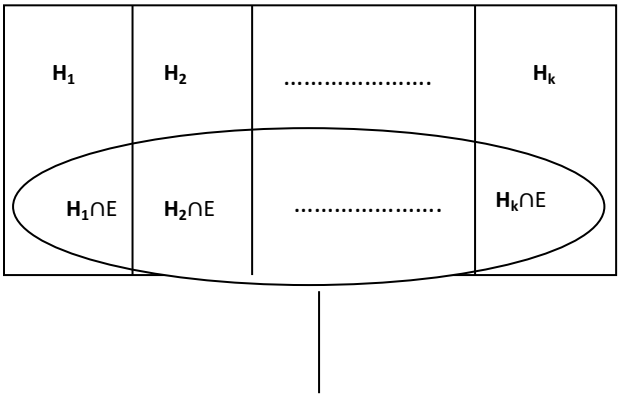
G / A	G	$\bar{G}$	Total
A	25	5	30
$\bar{A}$	15	5	20
Total	40	10	50

i)$P(A \cup G) = P(A) + P(G) - P(A \cap G) = \frac{30}{50} + \frac{40}{50} - \frac{25}{50} = \frac{45}{50}$

ii)$P(\bar{A}/G) = \frac{P(\bar{A} \cap G)}{P(G)} = \frac{15/50}{40/50} = \frac{3}{8}$

Baye's Theorem:

Let, S is the sample space having n equally likely outcomes. With some of outcomes let us define an event E. With some of outcomes of E we can define, separate mutually exclusive events $H_1, H_2, \ldots\ldots. H_k$. Then-



We have, $E = H_1 \cap E + H_2 \cap E + \ldots\ldots. + H_k \cap E$

$\therefore P(E) = P(H_1 \cap E) + P(H_2 \cap E) + \ldots\ldots + P(H_k \cap E)$

Again, $P(E/H_i) = \frac{P(H_i \cap E)}{P(H_i)}$; i = 1, 2, ……., k $\therefore P(H_i \cap E) = P(H_i)\, P(E/H_i)$

Now, Bayes' theorem states that, $P(H_i/E) = \frac{P(H_i \cap E)}{P(E)}$; i = 1, 2, ……., k

Example 9: In a box there are 70% mathematics books and 30% electrical engineering books. Among mathematics books 40% are foreign books and among electrical engineering books 50% are foreign books. A foreign book is selected. What is the probability that the selected one is an electrical engineering book?

Solution: Let, E: Foreign book H_1: Mathematics book H_2: Electrical engineering book

Given, $P(H_1) = 0.7$, $P(E|H_1) = 0.4$; $P(H_1 \cap E) = P(H_1)\, P(E|H_1) = 0.7 \times 0.4 = 0.28$

$P(H_2) = 0.3$, $P(E|H_2) = 0.5$; $P(H_2 \cap E) = P(H_2)\, P(E|H_2) = 0.3 \times 0.5 = 0.15$

$P(E) = P(H_1 \cap E) + P(H_2 \cap E) = 0.28 + 0.15 = 0.43$

So, $P(H_2|E) = \frac{P(H_2 \cap E)}{P(E)} = \frac{0.15}{0.43} = \frac{15}{43}$

Sample questions related to this chapter:

(1) Two robots are bought to work together in a restaurant to perform three tasks: Welcoming (W), Order placing (O) and Kebab making (K).
(i) Construct the sample space for assigning duties to two robots
(ii) Compute the probability that at least one robot will be making kebab
(iii) Compute the probability that the second robot makes kebab given the first robot welcomes the customers.

(2) Two robots can perform three tasks welcoming customers (W), singing (S) and dancing (D). The robotic restaurant owner assigns different tasks to different robots, as a result both robot can not perform the same duty.
(i) Construct the sample space for assigning duties to these robots
(ii) Compute the probability of second robot dances given the first robot sings.

(3) Signals are sent from Station-1 and Station-2. Fifty signals from Station-1 and 30 signals from Station-2 are sent daily. Among these signals 20% sent from Station-1 and 30% sent from Station-2 do not reach properly. In a random investigation it is found that one signal is not reached properly. Find the probability that the signal is sent from Station-2.

(4) Let us assume that 49% of adults in a Dhaka city area are male. One survey revealed that 75% female and 60% males are active Facebook users. From a Facebook users database, one person is selected at random. What is the probability that the selected Facebook user is a male?

(5) In a box there are 30 bulbs. The bulbs are identified by identity number 1 to 30. One bulb is selected at random. Find the probability that the number of selected bulb has the identity number **(i)** either multiple of 3 or 5, **(ii)** either multiple of 5 or 7, **(iii)** even under the condition that it is multiple of 3.

(6) In a mobile operator's office there are 5 electronic engineers and 6 computers scientists. A committee of 4 is to be formed to perform a special duty. Find the probability that the committee will consist of

i) all electronic engineers

ii) 2 electronic engineers and 2 computer scientists.